VISUALLY-VERIFIED AUTONOMOUS REWRITING

Viver: Visual Verification for Trusting Autonomous UI Rewrites

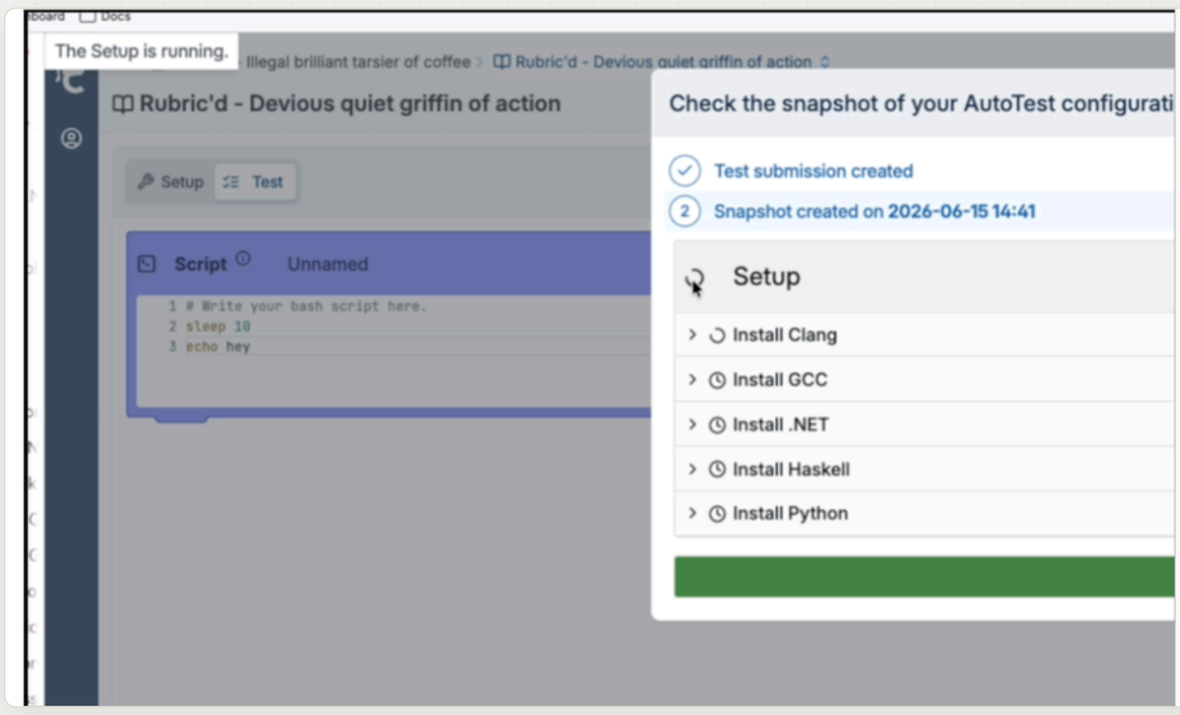
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CodeGrade

THE EXPERIMENT

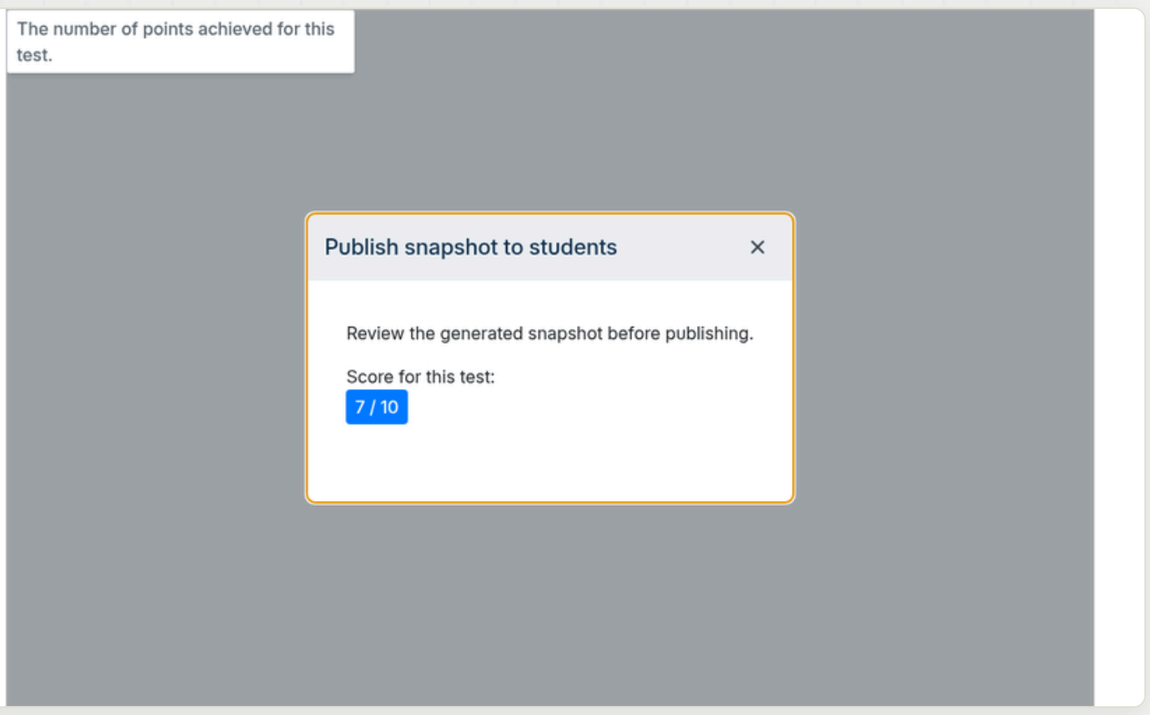
We gave Claude eyes.

And it shared its vision.

Claude spots the UI bugs a human reviewer catches, so it can **vet its own rewrites**.



REAL · TEST DEPLOY The migrated tooltip **detached to the top-left corner** while a modal was open. Every test passed. A human reviewer caught it (PROD-3516).



REPRODUCTION · GATE Rebuilt with the real components and shown to a blind, reference-free critic. ✓ CAUGHT 2/2 @ 0.88, in the full app. A component-only check is blind to it.

THE PROBLEM

A behaviour-preserving 1:1 dependency rewrite can pass every unit and component test while the rendered UI is visibly broken. The only thing that caught PROD-3516 was a reviewer's eye. To trust the rewrite unattended, the agent needs that eye.

THE PIPELINE TODAY · LEVEL 2

`read ticket` → `freeze tests` → `1:1 rewrite` → `tests green` → **`visual gate`** → `open PR`

A human still hands over each ticket. Level 3 is the same loop, but the agent spots the tech-debt ticket itself. The human steers, instead of driving.

FOUR QUESTIONS, FIVE SKILLS

RQ1 capture: render changed UI to an image, light and dark.

RQ2 reference-free: judge one screenshot, no before.

RQ3 differential: before/after, regression or intended?

RQ4 autonomy: compose them into a trustworthy rewrite.

DOES IT WORK?

RQ2 · REFERENCE-FREE

93% · 100%

recall and precision over 42 images. Catches 28 of 30 broken screens; every flag is real (0/12 false alarms).

RQ3 · BEFORE / AFTER

38 / 38

100% three-way. With the before as a spec, it tells a regression from an intended redesign.

RQ4 · ONE REAL MIGRATION

7/7 · 0

tests green, zero unplanned interventions. Selector-only test diff, ~35 min (PR #3814).

WHERE IT SLIPS

0.50 · 48px

a subtle low-contrast regression rated only coin-flip; a 48px panel shift read as intentional.

THE FINDING

Reference-free, Claude detects. Given the before, it judges intent.

A single screenshot reliably catches broken UI (93% / 100%), but it can read an intended change as a defect. The **before/after baseline is what supplies intent** (100% over 38 pairs). Honest limits: the diff set is not yet adversarial, a fully-hidden element still needs a functional check, and the human keeps the design gate.

`inspect-component-ui` `pr-screenshots` `visual-critic` `visual-diff-critic` `rewrite-dependency`

Five composable Claude Code skills, a blind-subagent eval harness, and a nine-type fault catalogue. Run them on any Cypress suite.

Code · results · shots
github.com/CodeGra-de/CodeGra.de/pull/3814
branch hackaton/claude-eyes